Components

1. Airframe & Propulsion

Component / item	Qty. / status	Role / where it works
5–6 inch carbon-fibre cinewhoop frame	1	Mechanical structure; carries battery, flight controller, compute and sensors.
Full propeller guards / ducts	1 set	Physically protects every propeller operating area, as required for safety.
Landing feet / landing skids	1 set	Supports take-off and landing and protects the underside.
2207-class brushless motors (2300–2450 KV, 4S compatible)	4	Convert electrical power into thrust; verify thrust with the final guarded frame.
5 × 3 or 5 × 3.1 two-blade propellers, matched to motors	At least 2 sets + spares	Generate lift and control; final prop size must match frame, guards and motor.
Motor mounting screws, frame fasteners and locknuts	1 kit	Secure the motors, guards and airframe.
Vibration-isolation mounts / damping material	As needed	Reduce vibration transferred to LiDAR, camera and flight controller.

2. Flight Control & Vehicle State

Component / item	Qty. / status	Role / where it works
Holybro Pixhawk 6C flight controller	1	Runs PX4 low-level flight control, attitude stabilisation, motor outputs and failsafes.
Compatible power module with voltage/current sensing	1	Feeds/monitors flight-controller power and provides battery measurements for failsafes.
Flight-controller mounting plate / vibration isolation	1 set	Mounts the FC securely and limits vibration.
Safety switch / arming indicator / buzzer, compatible with FC	As required	Supports safe arming, status indication and configuration.
MicroSD card / supported logging media	1	Stores flight logs and configuration data, if required by the FC.

3. Companion Compute & Storage

Component / item	Qty. / status	Role / where it works
NVIDIA Jetson Orin Nano Super 8GB + compatible carrier	1	Runs ROS 2, mapping integration, autonomous exploration, survivor detection and mission logic.
Compatible cooling solution (heatsink/fan) and airflow clearance	1 set	Prevents compute throttling or shutdown under sustained onboard processing.

Component / item	Qty. / status	Role / where it works
128GB high-endurance NVMe SSD + compatible adapter/slot	1	Stores OS, ROS packages, maps, logs and mission data.
Secure compute mounting plate, standoffs and fasteners	1 set	Keeps compute hardware secured during flight.

4. Mapping, Vision & Proximity Sensors

Component / item	Qty. / status	Role / where it works
Slamtec RPLIDAR A2M12 2D LiDAR	1	Provides 360° planar scans for indoor mapping, localisation and corridor/wall detection.
Intel RealSense D435i RGB-D camera	1	Provides RGB and depth for survivor perception and near-field obstacle checks; mount forward-facing.
ST VL53L1X time-of-flight distance sensors	2 recommended	Short-range distance sensing; use downward for height/landing and/or another direction after testing.
LiDAR mounting bracket with level adjustment	1	Holds scan plane level and protects the LiDAR from vibration/impact.
Camera bracket with tilt adjustment	1	Aims the camera at the useful forward field of view; calibrate camera-to-body transform.
Sensor cables and strain relief	1 set	Connects sensors reliably and prevents plugs/cables from moving or snagging.
Optional upward/side proximity sensor(s)	Only if testing proves needed	Adds coverage for ceiling or blind-side hazards; choose after arena tests, not by default.
Optional low-light/global-shutter camera alternative	Alternative only	Fallback if D435i is unavailable; depth and obstacle coverage would need a separate design.

5. Power, Battery & Charging

Component / item	Qty. / status	Role / where it works
4S 2200mAh high-discharge LiPo battery	1 onboard + 1 spare	Powers propulsion and onboard systems; confirm discharge rating against measured current.
LiPo balance charger compatible with 4S packs	1	Charges and balances flight batteries safely between tests.
Fire-resistant LiPo charging/storage bag or approved safe storage	1	Reduces risk during charging and storage; follow battery safety procedures.
Battery straps / anti-slip battery pad	1 set	Secures the battery and prevents it shifting in flight.
Power distribution board or rated power harness	1	Distributes battery power to ESC and required power branches.
5V high-current buck regulator (research suggests 8A rail)	1, subject to power audit	Provides a regulated supply for loads that require this rail; verify actual Jetson/carrier input requirements.
Clean 5V regulator rail for sensors/FC peripherals (research suggests 3A)	1, subject to power audit	Provides a separate low-noise supply for sensitive peripherals where required.
Rated connectors, silicone wire, heat-shrink and cable ties	1 kit	Creates safe, strain-relieved power and signal wiring.
Correctly rated fuse(s) / power protection	As required by final circuit	Protects wiring and branches from overcurrent; select after current and wire-gauge calculations.

Component / item	Qty. / status	Role / where it works
Battery voltage/current monitoring and low-battery thresholds	1 integrated system	Provides trustworthy battery state for return/abort failsafes.

6. Communication & Ground Control Station (GCS)

Component / item	Qty. / status	Role / where it works
Local telemetry radio/link with antennas, compatible with team setup	1 onboard set + GCS end	Carries telemetry and mission status over a local link; no public internet/cloud reliance.
Wi-Fi/local network adapter or router, if needed by chosen architecture	As required	Provides local connection for map, camera and ROS/GCS data; test range and interference.
Ground Control Station laptop	1	Runs monitoring, mission status, map and live camera display.
Telemetry receiver / USB adapter / required interface cables	1 GCS set	Connects the ground computer to the drone telemetry link.
Local network configuration and link-status monitoring	1 setup	Supports the required local communication and detection of link loss.
Foxglove Studio or RViz 2 display setup	1 software setup	Displays the live 2D map, drone pose/estimated position, camera feed, survivor tags and mission status.

7. Safety & Recovery

Component / item	Qty. / status	Role / where it works
Hardware emergency-stop / motor-enable interrupt circuit	1	Provides a physical safety stop; must be tested to cut motor-enable/arming authority as designed.
Mission-abort / emergency-recall control path	1	Lets the operator trigger permitted abort/recall from the GCS.
Communication-loss failsafe configuration	1 tested configuration	Defines safe onboard behaviour when GCS telemetry is lost; autonomy must not wait indefinitely.
Low-battery return/abort failsafe	1 tested configuration	Preserves enough reserve to return and land safely.
Geofence / arena-boundary failsafe	1 tested configuration	Prevents unintended movement beyond the allowed mission area.
Companion-computer heartbeat watchdog	1 software/hardware integration	Detects Jetson/ROS failure and ensures PX4 remains able to stabilise the vehicle.
Propeller-guard inspection and pre-flight checklist	1 checklist	Confirms guards, fasteners, battery, sensors and safety functions before each flight.
Suitable fire extinguisher / first-aid and test-area safety kit	Team/site provision	Supports safe bench, battery and flight testing; follow venue rules.

8. Wiring, Integration & Workshop Items

Component / item	Qty. / status	Role / where it works
USB 3 cable: RealSense to Jetson	1 compatible cable	Carries RGB-D data; use a secure, short cable with suitable bandwidth.
USB or UART cable: RPLIDAR to Jetson	1 compatible cable	Carries LiDAR scan data; interface depends on exact LiDAR/carrier configuration.

Component / item	Qty. / status	Role / where it works
UART MAVLink link: Jetson to Pixhawk	1 link + cable	Carries high-level velocity/position setpoints and telemetry; keep low-level motor control on PX4.
ESC signal/power connections (PWM or supported DShot)	4 motor channels	Connects PX4 motor outputs to ESC; verify firmware and ESC protocol compatibility.
I2C wiring/interface for ToF sensors	As required	Carries ToF measurements; check voltage levels, addresses and bus layout.
Level shifter / USB-UART adapter / connector adapters	Only if interface needs them	Bridges electrical/interface mismatches; use only after checking signal voltage and pinout.
Mounting hardware, standoffs, washers, thread-lock where appropriate	1 kit	Secures assemblies and prevents fasteners loosening.
Soldering and rework supplies, multimeter, current-limited bench supply	Team workshop kit	Enables safe assembly, continuity checks and staged power-up.
Spare propellers, screws, cable ties and connectors	1 spare kit	Reduces downtime after minor damage or wiring rework.

9. Software & Autonomy Stack (not physical components)

Component / item	Qty. / status	Role / where it works
PX4 Autopilot firmware	1 configuration	Low-level stabilisation, motor control, state estimation integration and flight failsafes.
ROS 2 Humble/Jazzy on supported Ubuntu version	1 software stack	Middleware for sensors, mapping, perception, planning and mission orchestration; validate ROS/Ubuntu compatibility.
MAVLink + PX4/ROS 2 bridge (e.g. supported MAVROS or px4_ros_com path)	1 integration	Connects companion computer to flight controller for telemetry and high-level commands.
slam_toolbox or validated 2D SLAM pipeline	1	Builds and updates the real-time 2D map from LiDAR and odometry.
Odometry, TF2 frames and sensor calibration	1 integration	Aligns map, body, LiDAR, camera and sensor coordinate frames.
Autonomous exploration/frontier selection + A* or equivalent planner	1 mission stack	Selects unexplored regions and computes collision-aware paths.
Obstacle detection and collision-avoidance logic	1 mission stack	Uses LiDAR/depth/proximity data to avoid walls, ceiling and other obstacles.
Survivor detector (lightweight vision model) + target dataset	1 model/pipeline	Detects survivor targets onboard and produces location estimates.
Survivor localisation, duplicate suppression and automatic map tagging	1 mission stack	Converts detections into map/grid coordinates and tags them autonomously.
Return-to-entry / return-path and landing logic	1 mission stack	Guides the drone back through the designated access point and lands.
GCS mission dashboard and local data link	1 application/setup	Displays live map, camera, estimated pose, survivor tags, battery, link status and mission progress.
Logging, replay and test scripts	1 setup	Supports repeatable bench, simulation and arena testing.